A standard model of arithmetic has only one isomorphism from the standard numbers

The Open Logic Project contributors

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Editorial note. The counted statement is the author's proposition environment labelled prop:thq-unique-iso (1323-1326) and the counted proof is the proof environment beginning at 1328, so the pairing is by adjacency.

Editorial note. The counted statement names the map g defined in the preceeding proposition and the hypothesis that M is standard, so that proposition's statement and the author's explanation of g are retained as context.

Editorial note. The counted proof is complete as the author wrote it and invokes only the definition of g, the homomorphism property of h and induction; it contains no citation command, figure, included file or non-ASCII character.

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit model-theory/models-of-arithmetic; no statement, proof step or symbol was rewritten, reordered or completed.

Editorial scope. One original result of the Open Logic Project unit model-theory/models-of-arithmetic, counted once: a standard model of arithmetic has exactly one isomorphism from the standard numbers. Only the counted statement, its own author proof and the author's statement of the preceding proposition defining the map are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- The map g. For a structure M satisfying Robinson arithmetic whose domain consists of the values of the numerals, g is the function from the standard numbers to the domain sending n to the value of the numeral n.. The counted statement names g, and the proof compares g with an arbitrary isomorphism pointwise (1329-1333).
- Isomorphism of structures. An isomorphism is a bijection between the domains that preserves the interpretations of the constant, the successor, addition, multiplication and the order relation.. The step of the induction uses that h commutes with the successor function, h of the successor of n equals the successor of h(n) (1343).
- Values of terms and numerals. The value of the numeral n+1 in M is the value of the term obtained by applying the successor symbol to the numeral n, hence is the successor in M of the value of the numeral n.. The

middle of the chain of identities rewrites the value of the numeral n+1 as the successor of the value of the numeral n (1337-1341).

Proposition 0.1. *If* $M \models \mathrm{Th}Q$*, and* $|\mathfrak{A}|M = \{\, \mathrm{val}(\underline{n})M \mid n \in \mathbb{N} \,\}$*, then* $\mathfrak{A}M$ *is standard.*

The function g is the most obvious way of defining a mapping from $\mathbb{N}$ to the domain of any other !!structure $\mathfrak{A}M$ for $\mathcal{L}L_A$, since every such $\mathfrak{A}M$ contains !!elements named by $\underline{0}$, $\underline{1}$, $\underline{2}$, etc. So it isn't surprising that if $\mathfrak{A}M$ makes at least some basic statements about the $\underline{n}$'s true in the same way that $\mathfrak{A}N$ does, and g is also bijective, then g will turn into an isomorphism. In fact, if $|\mathfrak{A}|M$ contains no !!elements other than what the $\underline{n}$'s name, it's the only one.

Proposition 0.2. *If* $\mathfrak{A}M$ *is standard, then* g *from the proof of 0.1 is the only isomorphism from* $\mathfrak{A}N$ *to* $\mathfrak{A}M$*.*

Proof. Suppose $h\colon \mathbb{N} \to |\mathfrak{A}|M$ is an isomorphism between $\mathfrak{A}N$ and $\mathfrak{A}M$. We show that $g = h$ by induction on n. If $n = 0$, then $g(0) = s0M$ by definition of g. But since h is an isomorphism, $h(0) = h(s0N) = s0M$, so $g(0) = h(0)$.

Now consider the case for $n + 1$. We have

$$\begin{aligned}
g(n+1) &= \mathrm{val}(\underline{n+1})M \text{ by definition of } g \\
&= \mathrm{val}(\underline{n}')M \text{ since } \underline{n+1}\,\mathrm{id}\,\underline{n}' \\
&= s\prime M(\mathrm{val}(\underline{n})M) \text{ by definition of } \mathrm{val}(t')M \\
&= s\prime M(g(n)) \text{ by definition of } g \\
&= s\prime M(h(n)) \text{ by induction hypothesis} \\
&= h(s\prime N(n)) \text{ since } h \text{ is an isomorphism} \\
&= h(n+1)
\end{aligned}$$

□